

- 6 It is given that $y = x^3 + px^2 + qx + 10$ where p and q are integers. The only values of x for which y is a decreasing function of x are those values for which $3 < x < 7$. Find the value of p and of q . [4]

For decreasing function,

$$\frac{dy}{dx} < 0$$

$$3x^2 + 2px + q < 0 \quad (M1)$$

Also $\Rightarrow 3 < x < 7$ (given)

$$\begin{aligned} \Rightarrow \frac{dy}{dx} &= 3(x-3)(x-7) \\ &= 3[x^2 - 7x - 3x + 21] \\ &= 3[x^2 - 10x + 21] \\ &= 3x^2 - 30x + 63 \quad (M1) \end{aligned}$$

By comparing,

$$\begin{aligned} 2p &= -30 \\ p &= -15 \quad (A1) \end{aligned}$$

$$\& \quad q = 63 \quad (A1)$$